Number Systems- Assignment

Q1. Complete the table with equivalents in Binary, Octal, and Hexa for: 143.75 (decimal), 11011.101 (binary), 453 (octal), 1F.C (hex).

Decimal	Binary	Octal	Hexa
143.75			
	11011.101		
		453	
			1F.C

Q2. Find equivalents:

a) $324_5 = ?_{10}$

b) $32.23_4 = ?_{10}$

c) $231_{10} = ?_7$

d) $189_{10} = ?_6$

Q3. Find the radix r:

a) $(312/20)_r = (13.1)_r$

b) $264_r + 136_r = 433_r$

c) $221505_8 = 12345_r$

Q4. Addition:

a) $1101_2 + 11011111_2$

b) $77652_8 + 76543_8$

c) $578A_{16} + ABDE_{16}$

d) $234_6 + 315_6$

Q5. Binary → Gray code:

a) 10101_2

b) 01110_2

Q6. Gray code → Binary:

a) 01101 (gray)

b) 10101 (gray)